

Stony Brook University
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Fall 2016

ASSIGNMENT 1

Due: Tuesday, September 20 in class.

INSTRUCTIONS: Show all your work or points will be deducted.

1. True—False—Uncertain:

- (a) A person who gives money away to people on the street does not have preferences that can be represented by a utility function.
- (b) If Marcos' utility function is $U(x, y) = xy$ and Laura's utility function is $U(x, y) = \sqrt{xy}$, then Marcos will always derive more happiness than Laura does from any bundle of x and y .
- (c) A monotonic transformation of a utility function does not change the marginal rate of substitution at any point.
- (d) If Brenda has downward-sloping indifference curves, then this is the same thing as saying that she prefers averages to extremes.
- (e) If Camilo is maximizing $U(x, y)$ subject to a budget constraint, then at the maximal point, the marginal utilities of x and y will be equal.

2. A review of Calculus I. Consider the real-valued function f defined over the interval $[-3, 3]$ by

$$f(x) = \begin{cases} 4 - (x + 2)^2 & \text{if } x \leq -1 \\ x^2(2 - x) & \text{if } x \geq -1 \end{cases}$$

- (a) Sketch a rough graph of the function.
 - (b) Find all critical points (i.e., points with zero first derivative) of the f function. Identify the interior local maxima and minima.
 - (c) Is the function discontinuous anywhere? Is the function non-differentiable anywhere? (Hint: to assess differentiability, check the graph, do you see any *kink* in the function?) If your answer is yes, is there a local maximum or minimum at these points?
 - (d) Does the function have any local maxima or minima at its end-points?
 - (e) Find its global maximum and minimum.
3. One way to show convexity of the indifference curves is to show that, for any two points (x_1, y_1) and (x_2, y_2) on an indifference curve that promises $U = k$, the utility associated with the point $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ is at least as great as k . Use this approach to discuss the convexity of the indifference curves for the following three functions. Graph your results.
- (a) $U(x, y) = \min(x, y)$.
 - (b) $U(x, y) = \max(x, y)$.
 - (c) $U(x, y) = x + y$.

4. Many advertising slogans seem to be asserting something about people's preferences. How would you capture the following slogans with a mathematical statement about the utility function? Either find a utility function for each of the examples or discuss what properties should it have.
- (a) Promise margarine is just as good as butter.
 - (b) Things go better with Coke.
 - (c) You can't eat just one Pringle's potato chip.
 - (d) Krispy Kreme glazed doughnuts are just better than Dunkin' Donuts.
5. Each day Paul, who is in third grade, eats lunch at school. He likes only Twinkies (t) and soda (s), and these provide him a utility of

$$U(t, s) = \sqrt{ts}.$$

- (a) If Twinkies cost \$0.10 each and soda costs \$0.25 per cup, how should Paul spend the \$1 his mother gives him to maximize his utility? Provide the levels of Twinkies and Soda that maximize Paul's utility and graph your answer.
 - (b) Suppose that the school tries to discourage Twinkie consumption by increasing the price to \$0.40. What will be the new levels of Twinkies and Soda that maximize Paul's utility after the price change? Graph your answer in the same diagram as in part a.
6. In this exercise, you will explore a preference relation and the utility that represents it that incorporates a *satiation point*. A satiation point is a bundle that provides complete "bliss" for the consumer. Suppose the bliss point is (x^*, y^*) . The ranking of all other bundles is determined by how close they are to (x^*, y^*) . Thus we have

$$(x_1, y_1) \succeq (x_2, y_2) \Leftrightarrow \sqrt{(x_1 - x^*)^2 + (y_1 - y^*)^2} \leq \sqrt{(x_2 - x^*)^2 + (y_2 - y^*)^2}$$

- (a) Graph some indifference curves for this preference relation. (Hint: A circle with center (x^*, y^*) and radius k is defined by the equation $(x - x^*)^2 + (y - y^*)^2 = k^2$.)
- (b) Suppose the consumer faced a budget constraint $p_x x + p_y y \leq I$. Does it necessarily have to hold that at a utility maximizing bundle we must have $\frac{p_x}{p_y} = MRS$? Explain your answer. (Hint: Consider the case in which the bliss point (x^*, y^*) is affordable?)